







# STATE TECHNICAL UNIVERSITY

END SEMESTER EXAMINATION 2025-2026 | ANSWER BOOKLET

CANDIDATE ROLL NO: 2026-CS-PH-006  
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EXAMINATION / COURSE: B.Tech Semester II  
SUBJECT & CODE: Physics (CS 201)  
BOOKLET SERIAL NO: BK-PH-80006

BARCODE SCAN TRACKING



\*BK-PH-80006\*

| Q.No  | Q1 | Q2 | Q3 | Q4 | Q5 |
|-------|----|----|----|----|----|
| Max   | 14 | 14 | 14 | 14 | 14 |
| Marks |    |    |    |    |    |

START WRITING YOUR ANSWERS BELOW. DO NOT WRITE IN THE MARGINS.

**Q.1** Ans I. De Broglie Waves - Group Velocity:

1. Matter waves have wavelength  $\lambda = h / mv$ .
2. Phase velocity  $v_p$  is the velocity of individual crests:  $v_p = \omega/k = E/p$ .
3. Group velocity  $v_g$  is velocity of packet =  $d(\omega)/dk$ .
4. Relation: For kinetic energy  $E = p^2/(2m)$ ,  $v_g = dE/dp = p/m = v$ .

Thus group velocity is equal to the particle velocity. This confirms that the particle moves with the center of the wave packet.







**Q.2**

Ans 2. Interference of Light:

Interference occurs when two coherent waves meet.

- Conditions: Coherent sources, monochromatic light, narrow slit sources.
- Division of Wavefront: Splits wavefront using slits (e.g. Young's Double Slit).
- Division of Amplitude: Splits amplitude by partial reflection (e.g. Newton's Rings, Thin films).
- In Thin films, path difference  $\Delta = 2 \mu t \cos(r) - \lambda/2$ .
- For constructive interference:  $2 \mu t \cos(r) = (2n-1) \lambda / 2$ .
- For destructive interference:  $2 \mu t \cos(r) = n \lambda$ .







**Q.3****Ans 3. Lasers & Optical Fibres:****1. Laser Basics:**

- Stimulated emission produces coherent photons.
- Population inversion is necessary where excited atoms exceed ground atoms ( $N_2 > N_1$ ).
- Main components: Active medium, Pumping source, Optical cavity.

**2. Optical Fibres:**

- Work on Total Internal Reflection (TIR).
- Core refractive index  $n_1$  is higher than Cladding  $n_2$ .
- Numerical Aperture  $NA = \sqrt{n_1^2 - n_2^2}$ .
- Acceptance angle  $\theta_a = \arcsin(NA)$ .







**Q.4**

Ans 4. Maxwell's Equations:

Maxwell unified electricity and magnetism into four equations:

1.  $\text{div } \mathbf{D} = \rho$  (Electric charges produce electric flux)
2.  $\text{div } \mathbf{B} = 0$  (No magnetic monopoles)
3.  $\text{curl } \mathbf{E} = - \frac{d\mathbf{B}}{dt}$  (Faraday's Law)
4.  $\text{curl } \mathbf{H} = \mathbf{J} + \frac{d\mathbf{D}}{dt}$  (Ampere-Maxwell Law with displacement current  $\mathbf{J}_d = \frac{d\mathbf{D}}{dt}$ )

Boundary conditions:

- Tangential  $\mathbf{E}$  field is continuous across interface:  $\mathbf{E}_{1t} = \mathbf{E}_{2t}$ .
- Normal  $\mathbf{B}$  field is continuous:  $\mathbf{B}_{1n} = \mathbf{B}_{2n}$ .
- Normal  $\mathbf{D}$  field:  $\mathbf{D}_{1n} - \mathbf{D}_{2n} = \text{surface charge density}$ .
- Poynting vector  $\mathbf{S} = \mathbf{E} \times \mathbf{H}$  represents power density.







**Q.5**

Ans 5. Solid State Physics:

1. Crystal Structures:

- SC: 1 atom/cell, packing fraction = 52%.
- BCC: 2 atoms/cell, packing fraction = 68%.
- FCC: 4 atoms/cell, packing fraction = 74%.
- Miller indices (hkl) denote crystal planes.

2. Semiconductors:

- Intrinsic: Pure silicon with equal electrons and holes ( $n = p$ ).
- Extrinsic: Doped with impurities (n-type with donors, p-type with acceptors).
- Fermi level  $E_f$  is the highest filled energy level at 0 Kelvin.
- Direct bandgap emits photons (GaAs), indirect emits phonons/heat (Si).

--- End of Answer Booklet (All 5 questions attempted) ---

Candidate Sign: Amit